Masserstein: linear regression of mass spectra by optimal transport

Michał Aleksander Ciach[1,3], Błażej Miasojedow[1], Grzegorz Skoraczyński[1], Szymon Majewski[2], Michał Startek[1], Dirk Valkenborg[3], and Anna Gambin[1]

[1]Faculty of Mathematics, Informatics and Mechanics, University of Warsaw, Warsaw, Poland
[2]Institute of Mathematics, Polish Academy of Sciences, Warsaw, Poland
[3]Faculty of Sciences, Hasselt University, Diepenbeek, Belgium

Abstract

RATIONALE: The linear regression of mass spectra is a computational problem defined as fitting a linear combination of reference spectra to an experimental one. It is typically used to estimate the relative quantities of selected ions. In this work, we study this problem in an abstract setting, in order to develop new approaches applicable to a diverse range of experiments.

METHODS: In order to overcome the sensitivity of the ordinary least squares regression to measurement inaccuracies, we base our methods on a non-conventional spectral dissimilarity measure, known as the Wasserstein or the Earth Mover's distance. This distance is based on the notion of the cost of transporting signal between mass spectra, which renders it naturally robust to measurement inaccuracies in the mass domain.

RESULTS: Using a data set of 200 mass spectra, we demonstrate that our approach is capable of accurate estimation of ion proportions without extensive pre-processing of spectra required by other methods. The conclusions are further substantiated using data sets simulated in a way that mimics most of the measurement inaccuracies occurring in real experiments.

CONCLUSIONS: We have developed a linear regression algorithm based on the notion of the cost of transporting signal between spectra. Our implementation is available in a Python 3 package called `masserstein`, freely available at `https://github.com/mciach/masserstein`.

1 Introduction

A frequently encountered task in mass spectrometry is the quantification of signal corresponding to a particular set of ions. From an algorithmic point of view, the most challenging parts of this problem are the separation of the signal from the noise and the separation of overlapping isotopic envelopes. Numerous approaches have been developed in order to tackle this problem in the context of specific experiments. These include the estimation of reaction rates in ETD fragmentation [1]; quantification of polymer chain lengths and compositions [2, 3]; annotation of MS^2 spectra in data independent acquisition label-free quantification experiments [4]; studies of fragmentation of aliphatic diselenides and selenosulfenates [5, 6], and studies of protein deamidation and ^{18}O labelling [7].

Despite the abundance of different algorithmic approaches and software tools, a common theme is the approximation of an experimentally observed spectrum by a set of reference spectra. Therefore, from a mathematical point of view, all the described problems can be expressed with a single equation:

$$\mu = p_1\nu_1 + p_2\nu_2 + \cdots + p_k\nu_k.$$

Here, μ is the observed spectrum, ν_i's are the reference spectra of the ions in question, and p_i's are the unknown proportions of the latter. Since, in mass spectrometry, the reference spectra are

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1002/rcm.8956

most often predicted using computational methods, we refer to them as the *theoretical* spectra. By analogy to the ordinary linear regression known from the field of statistics, in this work we call the problem of finding p_i the *linear regression of mass spectra*.

The equation presented above has appeared e.g. in [4], where the authors have used it to annotate data independent acquisition label-free quantification experiments. Figure 1 illustrates a linear regression of a mass spectrum consisting of overlapping isotopic envelopes of human hemoglobin subunits α and δ.

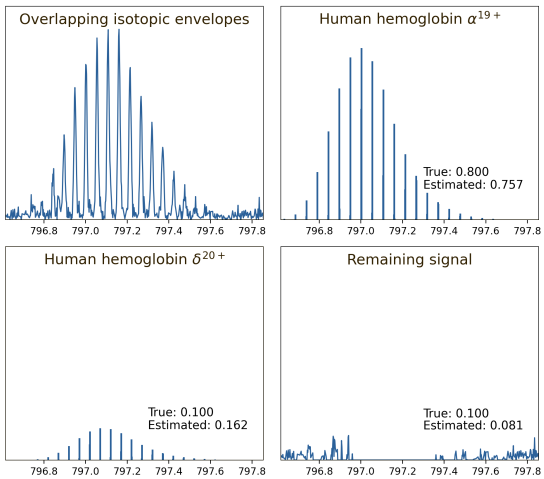

Figure 1: An example of a regression of a simulated human hemoglobin ESI MS^1 spectrum. Using the method presented in this article, we have separated the signal into α 19+ and δ 20+ subunits and the remaining background noise. The proportions were estimated directly from the top-left spectrum, without any additional preprocessing such as peak picking or denoising.

A common theme of different approaches to linear regression of mass spectra is the assumption that the reference spectra are known. Therefore, from the methodological point of view, this problem is an exact opposite of molecule identification, where the task is to identify the identity of ions, but not their quantities.

In the case when different isotopic envelopes do not overlap, the background noise is small, and there is only a handful of molecules of interest, linear regression of spectra can easily be performed manually by integrating selected regions of the experimental spectrum. However, algorithmic approaches need to be employed when there is a considerable overlap of isotopic envelopes, the signal-to-noise ratio is small, or thousands of molecules need to be analyzed in a high-throughput setting. The ever-increasing popularity of high-throughput methods, such as mass spectrometry imaging, calls for advanced algorithmic solutions which allow for rapid processing of massive data sets. In those types of experiments, and with software tools that neglect computational optimization, the time needed to process a single data set can take up to several days, needlessly extending the time needed to analyze the sample of interest.

Unfortunately, the currently used terminology is somewhat misleading. The problem of estimating abundances of overlapping isotopic envelopes is known in the literature under several names, most common ones being the *resolving of isobaric interferences* and *deconvolution*. Since the experimental spectrum is expressed as a linear combination of the theoretical ones, and the task here is to separate intermixed signals of different molecules, this problem has also been referred to as the *linear deconvolution* [4]. However, deconvolution has several meanings which have little in common. One of the most common ones is converting m/z values into masses, as exemplified by the popular MaxEnt algorithm [8], also referred to as the *charge deconvolution*. We therefore introduce the term *linear regression of mass spectra* in order to avoid confusion with other algorithmic problems of mass spectrometry.

A common approach to the regression of mass spectra, found e.g. in the specter [4] and the `masstodon` [9] tools, is to perform an ordinary least squares regression where the experimental spectrum is treated as the dependent variable and the theoretical ones as independent variables. Mathematically, this technique minimizes the Euclidean distance between the experimental spectrum and a linear combination of the theoretical ones. A similar technique, the L1 regression (also known as the least absolute deviation regression), is sometimes used [3]. Another approach is a sequential subtraction of estimated signal from the experimental spectrum, as exemplified by the THRASH algorithm [10].

The performance of such methods is hindered by a series of problems. Most of them arise from the fact that they are based on pointwise comparison of spectra, that is, they compare peaks with the same m/z values. However, unlike the theoretically predicted spectra, experimental ones have limited resolution and accuracy. Because of this, peak picking of the experimental spectrum is required, which is often imperfect (such as the conventional peak centroiding) or computationally expensive (such as the continuous wavelet transform approach [11]). The experimentally obtained peaks virtually never match the theoretical ones perfectly due to accuracy limits of the instrument and numerical errors of peak-picking procedures. This limits the performance and applicability of most of the approaches to measure the similarity between spectra, like the Euclidean distance, correlation, spectral contrast angle [12], or the entropy function [13].

Our contribution. The goal of this work is to further investigate the problem of linear regression of mass spectra. We develop a method that is robust against measurement inaccuracies and numerical errors of peak picking algorithms, while being resilient to interfering signals coming from chemical impurities and background noise.

The main novelty in our approach to linear regression is the use of the Wasserstein distance, which is a recently proposed approach to processing mass spectra [14]. This distance, based on the theory of the optimal transport, is naturally robust to uncertainties in m/z measurements and different resolutions of the compared spectra. In particular, to our knowledge, this is the first algorithm that is capable of explaining an experimental spectrum in profile mode using a set of computationally generated, infinitely resolved theoretical spectra. On the other hand, when the experimental spectrum is analyzed in centroid mode, our algorithm does not require peak matching, and is therefore capable of utilizing the full information in both the experimental and the theoretical spectra.

We tackle the problem of linear regression of mass spectra in its abstract form instead of focusing on a particular type of experiment. This way, while being mathematically sound, our methods retain their full scope of applicability, from analytical chemistry to metabolomics to proteomics to synthetic polymer science. They are not limited to any single type of mass spectrometer or pre-processing software. The mathematical foundations stay the same regardless whether an FTICR, TOF, or quadrupole instrument is used.

Accordingly, instead of presenting an application of our linear regression algorithm to any particular experiment, we test its performance on a custom-made data set consisting of 200 repeated measurements of the same set of compounds. This allows us to assess both the accuracy and variance of the estimation of ion proportions. We further confirm our results using simulated data sets, where we take into account several measurement inaccuracies occurring naturally in mass spectra.

The regression algorithm presented in this work has been implemented in the Python 3 language, and is freely available as a set of command-line applications and a Python 3 package `masserstein` at `https://github.com/mciach/masserstein`.

Structure of the article. First, we give an introduction on the Wasserstein distance, an optimal transport-based approach to the comparison of spectra. Next, we describe the problem of the linear regression of mass spectra in detail and derive a regression algorithm based on the Wasserstein distance, which is the main contribution of this work. Finally, we assess the performance of the algorithm on a set of 200 experimental spectra as well as on computationally generated datasets.

2 Methods

As described in the introduction, naturally occurring measurement inaccuracies hinder the performance of the common approaches to the estimation of relative abundances of ions in mass spectra. In order to overcome these problems, a spectrum dissimilarity measure based on the theory of optimal transport was recently investigated, with the aim to develop a linear regression method that allows to compare spectra with different resolutions and is robust to measurement errors in the mass domain [14]. The measure is known in the field of probability theory as the Wasserstein metric [15], and in the computer science community as the Earth Mover's distance [16].

The idea behind the measure is to transport the ion current from one spectrum onto the other one and quantify the distance that the current needs to travel. The dissimilarity between the two spectra is equal to the minimal distance in m/z domain that needs to be traveled in order to fully transform one spectrum into the other. In particular, the Wasserstein distance between MS^1 spectra of two ions with the same charge is approximately equal to the molecules' mass difference. This interpretation allows to consider as fairly similar those spectra for which the Wasserstein distance is less than one hydrogen mass.

Examples of the values of the Wasserstein distance between pairs of spectra are shown in Fig. 2. The spectra are artificially constructed in order to provide simple and clear examples. Consequently, we have purposefully neglected several important phenomena occurring in actual spectra, such as background noise, which will be dealt with later on in this work. Worked examples of how to compute this distance between pairs of spectra are also provided later on in this work.

In the right panel of Fig. 2, we consider a pair of corresponding spectra in profile and centroid mode. Even though such spectra are incomparable using conventional measures, such as the Euclidean distance, the Wasserstein distance between them has a small value of 0.032 Da. This value reflects the different resolutions of those spectra (where the centroid-mode spectrum is assumed to have an infinite resolution).

The left panel of Fig. 2 presents a pair of spectra with no matching peaks. Pointwise measures, such as correlation, do not capture any similarity between them (note that, in certain applications, this may be a desirable phenomenon). The Wasserstein distance, on the other hand, again has a small value of 0.078 Da, indicating that those spectra are very similar in terms of the m/z differences of peak positions. The fact that the peaks do not match, however, is still reflected by the Wasserstein distance, since it is over twice as large as between the spectra in the right panel. The exact method of calculation of this distance and some of its properties when applied to mass spectra are discussed further in the article.

An alternative interpretation of the Wasserstein distance is the minimal amount of distortion such as shifting, broadening and narrowing of peaks that is required to transform one of the spectra into the other. This approach is naturally robust to small distortions in the m/z or intensity measurements, and has no requirements as to the accuracy or resolution of the measurement. In particular, to the best of our knowledge, it is the only similarity measure capable of an accurate comparison of profile and centroided spectra.

In the following subsections, we give a short introduction to the mathematical formalism behind the Wasserstein distance and show several examples that illustrate its properties related to mass spectra. Next, we develop a regression algorithm which is the main contribution of this work. The algorithm improves upon the original one published in [14] by taking into account the experimental signal that does not correspond to the theoretical spectra. This makes the method robust to background noise, and makes it applicable to a broader class of experimental spectra. We also improve on the computational complexity of the original algorithm.

2.1 The Wasserstein distance

Let μ and ν be any two mass spectra to be compared. In order to simplify the description of the Wasserstein distance and the concept of transporting signal between spectra, we will assume for now that both spectra are centroided. The case of profile spectra will be discussed later on.

In order to compare μ and ν, we aim to transport all the signal from one spectrum to the other and quantify the minimal amount of distance in the mass domain over which the signal needs to be transported. We do not differentiate between a source and a target spectrum, so that the final distance is symmetrical—in other words, we will assume that transforming μ into ν inflicts the same cost as transforming ν into μ.

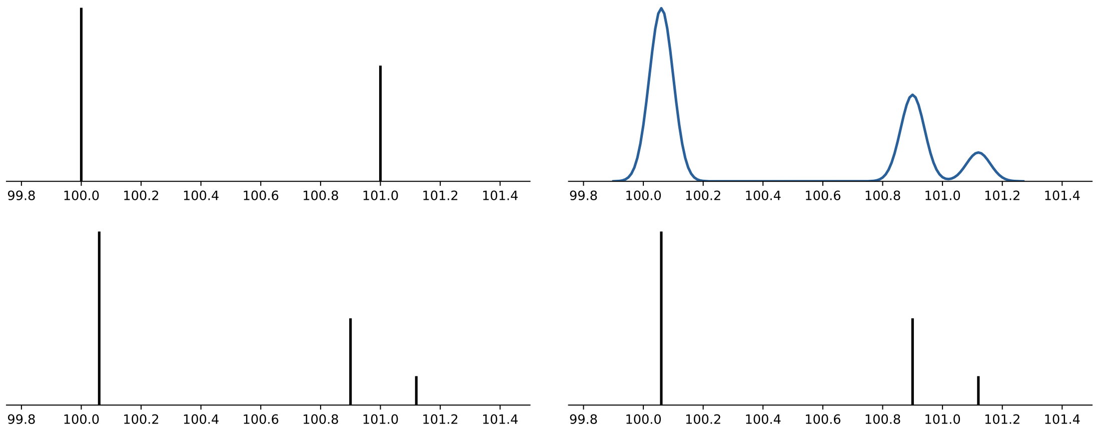

Figure 2: Example values of the Wasserstein distance between mass spectra. Left: Two spectra in centroid mode, with the Wasserstein distance between them being equal to 0.078 Da. Right: A spectrum in profile mode and a corresponding centroided one, with the Wasserstein distance equal to 0.032 Da. Both values, being less than 1 hydrogen mass, indicate a fairly high degree of similarity, even though no peaks match in the first example and the spectra are in different modes (profile vs centroid) in the second one.

We treat mass spectra as functions defined on the real line $\mathbb{R}$, so that $\mu(x)$ is the intensity at the m/z value x in the spectrum μ. We assume that all the considered spectra are normalized by their total ion current, so that the peak intensities of each spectrum (including the peaks arising from background noise or contaminants) sum up to 1. Note that such normalization may be meaningless from a data analyst's point of view, because the normalizing factor includes the noise intensity. However, it is often used for technical reasons when developing computational methods because it allows to treat mass spectra as probability measures and use the tools of probability theory to analyze them (see, e.g., [13]). We show how to circumvent this assumption in the Results section.

Let $\gamma(x, y)$ be the amount of signal transported between the point x in spectrum μ and the point y in spectrum ν. The function γ is referred to as a *transport plan*.

Any transport plan needs to satisfy the following properties:

$$\sum_{y \in \nu} \gamma(x, y) = \mu(x), \qquad \sum_{x \in \mu} \gamma(x, y) = \nu(y). \tag{1}$$

The first of those properties means that all the signal intensity $\mu(x)$ needs to be transported somewhere into the spectrum ν. Similarly, the second property means that the intensity $\nu(y)$ is fully filled by the ion current coming from the spectrum μ. Naturally, the transport plan also needs to be non-negative, $\gamma(x, y) \geq 0$, as we cannot transport negative intensity. In turn, any function satisfying the above properties is a valid transport plan. We denote by Γ the space of all possible transport plans.

The cost of a given transport plan is the total distance traveled by the ion current. This is calculated by multiplying the distance between the points x and y by the amount of signal traveling between them, and summing over all peaks in both spectra:

$$\sum_{x \in \mu, y \in \nu} |x - y| \gamma(x, y).$$

Note that the dependence of this sum on the spectra μ and ν is only implicitly expressed through the function γ and equations (1).

We assume that the transport is not directed, so that transporting the intensity in the direction of increasing mass inflicts the same cost as in the other direction. This is formally expressed by the absolute difference between the points in the above equation.

The Wasserstein distance [15, 17] between two spectra, $W(\mu, \nu)$, is defined as the minimal cost of transport over all possible transport plans from the space Γ:

$$W(\mu, \nu) = \min_{\gamma \in \Gamma} \sum_{x \in \mu, y \in \nu} |x - y| \gamma(x, y), \tag{2}$$

The function W defined this way satisfies the mathematical properties of a distance function [15], that is, non-negativity $W(\mu, \nu) \geq 0$, symmetry $W(\mu, \nu) = W(\nu, \mu)$ and the triangle inequality $W(\mu, \nu) \leq W(\mu, \zeta) + W(\zeta, \nu)$.

Since the distance is symmetric, there is no designated *source* spectrum of the transported signal, nor a *target* spectrum to which the signal is transported. However, when depicting transport plans for $W(\mu, \nu)$, we adopt a convention that the signal is transported from ν to μ.

Although the formulation of the distance may seem baffling, it turns out that under reasonable assumptions the algorithm to compute $W(\mu, \nu)$ has a linear time complexity [17]. The only requirement for this is that the input spectra are available as lists of m/z values and the corresponding signal intensities, sorted in the order of increasing mass. Since raw spectra, in both centroid and profile mode, are usually stored this way on computers, the Wasserstein metric is computationally equally efficient to the Jaccard score and the Euclidean distance.

The algorithm to compute the Wasserstein distance between a pair of spectra is based on the following theorem [17, 18]:

Theorem 1 *Let μ and ν be two probability measures on the real line $\mathbb{R}$. Let M and N be the cumulative distribution functions of μ and ν respectively. Then,*

$$W(\mu, \nu) = \int_{\mathbb{R}} \Big| M(t) - N(t) \Big| dt. \tag{3}$$

The above theorem can be applied to spectra normalized by their total ion current, since such spectra can be interpreted as probability measures. It captures both the case of centroided and profile spectra. In the case of the former, however, the cumulative distribution functions are step functions, and the formula can be considerably simplified.

Theorem 2 *Let μ, ν be two centroided mass spectra normalized by the total ion current. Let $S = \{s_1, s_2, \ldots, s_n\}$ be an ordered list of all distinct masses in both spectra. Let M and N be the cumulative distribution functions of μ and ν, i.e. $M(t) = \sum_{x \leq t} \mu(x)$. Then,*

$$W(\mu, \nu) = \sum_{i=1}^{n-1} (s_{i+1} - s_i) \Big| M(s_i) - N(s_i) \Big|. \tag{4}$$

Formula (4) admits a simple interpretation. Observe that $M(s_i) - N(s_i)$ is the difference in ion current on the left hand side of point s_i, which needs to be transferred either to or from the point s_{i+1} in order to achieve balance.

In order to illustrate the computation of the Wasserstein distance and make the concept more intuitive to the reader, we present two worked examples below.

Example 1. Consider two abstract spectra, μ and ν, such that μ is concentrated at 100 Da (i.e. $\mu(100.5) = 1$) and ν is distributed evenly over $98, 99, 100, 101$ and 102 Da (i.e. $\nu(98) = \nu(99) = \nu(100) = \nu(101) = \nu(102) = 0.2$). Obviously, those spectra do not correspond to any actual ions. They only serve for an easy illustration of the computation of the optimal signal transport. The procedure is illustrated in Fig. 3.

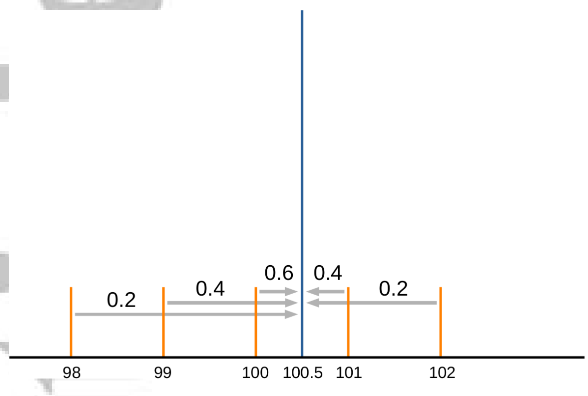

Figure 3: An example of an optimal transport scheme between two abstract spectra μ and ν, depicted on a single graph in blue and orange respectively. The signal of the spectrum μ is concentrated at the point 100.5, while ν is distributed evenly at five points from 98 to 102. Grey arrows depict the transport of the ion current. Numbers above arrows show the proportions of the ion current flowing between neighboring peaks, i.e. the absolute difference of the cumulative distribution functions.

The cumulative distribution function of μ and ν are given by

$$M(t) = \begin{cases} 0.0 & : \quad t < 100.5, \\ 1.0 & : \quad 100.5 \leq t, \end{cases}$$

and

$$N(t) = \begin{cases} 0.0 & : \quad t < 98, \\ 0.2 & : \quad 98 \leq t < 99, \\ 0.4 & : \quad 99 \leq t < 100, \\ 0.6 & : \quad 100 \leq t < 101, \\ 0.8 & : \quad 101 \leq t < 102, \\ 1.0 & : \quad 102 \leq t. \end{cases}$$

The list of all distinct masses, S, is now equal to $(98, 99, 100, 100.5, 101, 102)$, and the difference between the cumulative distribution functions, $N(t) - M(t)$, indicating the ion current imbalance at point t, is equal to

$$N(t) - M(t) = \begin{cases} 0.0 & : \quad t < 98.0, \\ 0.2 & : \quad 98.0 \leq t < 99.0, \\ 0.4 & : \quad 99.0 \leq t < 100.0, \\ 0.6 & : \quad 100.0 \leq t < 100.5, \\ -0.4 & : \quad 100.5 \leq t < 101.0, \\ -0.2 & : \quad 101 \leq t < 102, \\ 0.0 & : \quad 102 \leq t. \end{cases}$$

From the above equation, we can read out that 0.2 of the signal is transported from the point 98.0 to 99.0; 0.4 of the signal is transported from 99.0 to 100.0; 0.6 from 100.0 to 100.5. Next, as the sign of the imbalance changes, so does the direction of transport between neighboring points, and so 0.4 of the signal is transported from 101.0 to 100.5, and 0.2 of the signal from 102 to 101. Finally, the ion currents of both spectra balance out at the point 102.

The final distance can be computed by taking the absolute values of the ion current imbalance and multiplying them by the distance travelled, so that $W(\mu, \nu) = 0.2 \cdot 1 + 0.4 \cdot 1 + 0.6 \cdot 0.5 + 0.4 \cdot 0.5 + 0.2 \cdot 1 = 1.3$ Da.

Example 2. Consider a spectrum μ concentrated at 100 Da, and a profile spectrum ν consisting of a single Gaussian peak centered at 100 Da with a standard deviation σ. We therefore have $\mu(100) = 1$ and

$$\nu(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-100)^2}{2\sigma^2}}.$$

Let Φ be the cumulative distribution function of the standard Gaussian random variable:

$$\Phi(t) = \int_{x \leq t} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

The CDF of ν is then given by $N(t) = \Phi((t - 100)/\sigma)$. The exact value of the Wasserstein distance between μ and ν is then equal to

$$W(\mu, \nu) = \int |M(t) - N(t)| dt = \int_{x \leq 100} |N(t)| + \int_{x \geq 100} |N(t) - 1| = \int_{x \leq 100} N(t) dt + \int_{x \geq 100} (1 - N(t)) dt.$$

After plugging in Φ in the above integrals and substituting the variables, one arrives at

$$W(\mu, \nu) = \sigma \int_{x \le 0} \Phi(x)dx + \sigma \int_{x \ge 0} (1 - \Phi(x))dx.$$

Using the relation $\Phi(-x) = 1 - \Phi(x)$, we get

$$W(\mu, \nu) = \sigma \int_{x \le 0} \Phi(x)dx + \sigma \int_{x \ge 0} \Phi(-x)dx = 2\sigma \int_{x \le 0} \Phi(x)dx.$$

As the antiderivative of $\Phi(x)$ is $x\Phi(x) + \phi(x) + C$, where ϕ is the density of a standard Gaussian variable, we arrive at

$$W(\mu, \nu) = 2\sigma\phi(0) = 2\sigma \frac{1}{\sqrt{2\pi}} = \sigma\sqrt{\frac{2}{\pi}}.$$

Therefore, we obtain a simple formula for the distance between a centroided spectrum and a corresponding profile spectrum with Gaussian peaks of standard deviation σ. This result explains the small distance between spectra in the right panel of Fig. 2.

Handling profile spectra in practice. Although, in principle, profile spectra are continuous functions, they are usually represented as finite lists of mass and intensity pairs. It turns out that such lists can be simply treated as centroid spectra when the Wasserstein distance is computed. In that case, Theorem 2 gives an accurate approximation of the cost of the optimal transport plan.

Assume we are given a finite list of mass and intensity pairs, (s_i, I_i) for $i = 1, 2, \dots, n$, approximating a profile spectrum μ. Assume as well that we have a constant spacing between consecutive intensity measurements, so that $s_{i+1} - s_i = 1/n$ for $i = 1, 2, \dots, n-1$. Treating μ in the same way as a centroided spectrum, we compute its cumulative distribution function as $\hat{M}(t) = \sum_{x_i \le t} I_i / \sum I_i$. The true cumulative distribution function of μ, on the other hand, is given by $M(t) = \int_{x \le t} \mu(x)dx / \int \mu(x)dx$. Now, we have

$$\hat{M}(t) = \frac{\sum_{x_i \le t} I_i}{\sum I_i} = \frac{\frac{1}{n}\sum_{x_i \le t} I_i}{\frac{1}{n}\sum x_i I_i} \approx \frac{\int_{x \le t} \mu(x)dx}{\int \mu(x)dx} = M(t)$$

The assumption of an uniform signal sampling (i.e. a constant spacing between consecutive measurements) is crucial for the approximation to work. When this is not satisfied, a spectrum needs to be resampled prior to computing the distance. However, care needs to be taken to resample the spectrum in a way that does not lead to a loss of data or an introduction of additional noise signal. The resampled spectrum should therefore always be checked visually for the presence of any introduced artifacts, and the total ion current of the original and the resampled spectra should be compared.

For a practical example of this procedure, see Section 3. The resampling algorithm used in this study, based on a piecewise-linear interpolation of profile spectra, is presented in section S4 of the Appendix. We also note that further research on the Wasserstein distance between profile spectra sampled on not uniform m/z arrays would allow for a more rigorous validation of the resampling procedure by comparing the original and the resampled spectrum.

2.2 Linear regression of mass spectra

In this section, we formally define the problem of linear regression of mass spectra. We consider an experimentally observed spectrum, denoted μ, and a library of theortical spectra, denoted $\nu_1, \dots, \nu_k$. Although referred to as theoretical, the spectra may come from computational simulations, experimental measurements, or a combination of both.

As before, we assume that all spectra are normalized by their total ion current. Let $p = (p_1, p_2, \dots, p_k)$ be a vector of non-negative *weights* or *proportions* such that $p_1 + \dots + p_k \le 1$. Define ν_p to be a linear combination of the theoretical spectra with weights p, i.e.

$$\nu_p = p_1\nu_1 + \dots + p_k\nu_k.$$

We will call ν_p a *model spectrum*, in analogy to a model matrix used in an ordinary least squares linear regression. Note that the ν_p's intensity sums up to $p_1 + \cdots + p_k$, which may be less than 1. The inequality is allowed to account for the fact that ν_p may not explain all of μ's signal.

We assume that the observed spectrum can be approximated by the model spectrum with some additional chemical and/or background noise ε, so that for any m/z value x we have

$$\mu(x) = \nu_p(x) + \varepsilon(x). \tag{5}$$

Note that because of the assumed normalization of the spectra, the total signal of ν_p is the proportion of μ that is explained by the model.

In the most general setting, the problem of linear regression of mass spectra is defined as finding the proportions that minimize some chosen distance function d between the experimental and the model spectrum:

$$p^* = \arg\min_p d(\mu, \nu_p). \tag{6}$$

For example, in the commonly used least-squares approach, $d(\mu, \nu_p) = \sum_{x \in S}(\mu(x) - \nu_p(x))^2$, where S is the set of observed peak masses. As described in the introduction, this particular distance is sensitive to different resolutions and mass measurement errors, which makes it necessary to centroid the spectra and set a mass window for peak matching. The Wasserstein distance, on the other hand, is naturally robust to small errors in the mass measurement. Therefore, it is a natural candidate for a distance measure to be used for a linear regression of mass spectra.

Regression by optimal transport. If all the signal of μ can be explained by the model spectrum ν_p, the regression problem can be expressed as finding proportions p^* such that

$$p^* = \arg\min_p W(\mu, \nu_p).$$

This optimization problem has been shown to yield accurate results when the only differences between μ and ν_p are caused by mass measurement errors and differing resolution [14]. However, experimentally obtained spectra most often contain signals which are not theoretically predicted, like chemical contaminants or background noise. Such signals strongly disturb the optimal transport plan, leading to incorrect estimation of the proportions p^*. This is because the Wasserstein distance requires both spectra to have equal amounts of intensity, enforcing $p_1 + \cdots + p_k$ to be equal to 1.

To account for the additional signal in μ, we extend the m/z axis by adding an auxilliary point ω onto which such signal can be transported. All the signal transported onto ω is assumed to be unexplained by the model spectrum, and usually treated as noise that can be removed from μ. This applies regardless of the nature of the transported signal, i.e. whether it is a contaminant or background electronic noise. The idea behind this approach is visualized in Fig. 4.

We assume that the cost of transporting signal from μ to ω does not depend on the m/z value of the signal (equivalently, that ω is equidistant to all points on $\mathbb{R}$). We denote as κ the cost of transporting signal from μ to ω (equivalently, the distance between ω and any point on $\mathbb{R}$). Since κ is interpreted as the cost of removing noisy signal from μ, we refer to it as the *denoising penalty.*

The Wasserstein regression of mass spectra, accounting for signal unexplained by the model spectrum, is defined as

$$p^* = \arg\min_p W(\mu, p_0\omega + \nu_p), \tag{7}$$

where $p_0 = 1 - p_1 - \cdots - p_k$ is the amount of the unexplained signal in μ transported onto ω. During minimization of the Wasserstein distance above, the observed signal that cannot be feasibly transported onto any theoretical spectrum gets transported onto ω. An important property of this approach is that the decision whether to remove a given signal is performed independently for each intensity measurement in the experimental spectrum. This allows it to work properly in the case when the number of the background noise peaks greatly exceeds the number of the peaks of the molecules of interest, which is usually the case in mass spectrometry. For example, the noise peaks in the middle of Fig. 4 are going to be removed (i.e. transported into ω) regardless of their number.

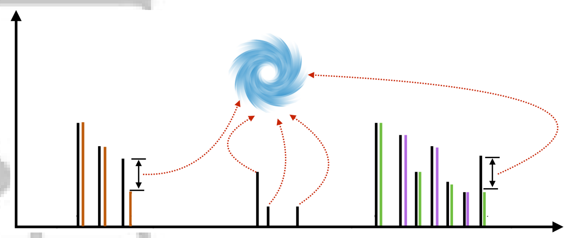

Figure 4: An illustration of the Wasserstein regression of mass spectra. An experimental spectrum (black) is explained by a set of three theoretical spectra (orange, green and magenta). Excess signal from the theoretical spectrum, occurring due to background noise or sample impurities, is transported onto an auxilliary point ω, represented as the vortex.

Note that the denoising penalty κ admits a physical interpretation in terms of m/z units. Transporting intensity from the experimental to the model spectrum over a distance larger than κ is more costly than removing it by transporting it to ω. The penalty can therefore be treated as a maximum distance over which the transport is feasible. This allows for some intuition behind the optimal values of this parameter: the maximum feasible transport distance should be set as the smallest value that allows to match corresponding theoretical and experimental peaks. The instrument accuracy (in Dalton units) is therefore an example of a reasonable value for κ. In practice, however, the choice of this parameter is more complicated. This is because, apart from the instrument accuracy, there are several factors that influence the distance between experimentally observed peaks and their theoretical counterparts, including, but not limited to, the resolving power. Therefore, usually the results for several different values of κ need to be inspected manually. This issue is discussed in more detail in the subsequent sections.

A major advantage of our approach, as opposed to matching peaks by mass windows in methods based on least squares regression, is that κ does not set a hard threshold on the transport distance, allowing for more flexibility in the choice in this parameter. Furthermore, in some cases it may be beneficial to transport the signal over distances larger than κ, while in other cases interfering signal is removed regardless of its proximity to theoretical peaks. Specifically, whether a given signal is removed depends not only on it's distance from the nearest theoretical peaks, but also on the shapes of the theoretical isotopic envelopes—in contrast to methods based on linear regression, in which a signal is always incorporated when it's sufficiently close to any theoretical peak. This phenomenon is illustrated by the following example.

Example 3. Consider a theoretical spectrum ν consisting of n peaks, and let the i-th peak be at m/z $m_1 + i/q$ for some $m_i \geq 0$, and let it have intensity a_i, where $a_1 + a_2 + \cdots + a_n = 1$. This models a low-resolution theoretical spectrum of an ion with charge q and with the monoisotopic peak at m_1. Since the proportions a_i are arbitrary, this model encompasses all possible single-ion low-resolution spectra, allowing us to obtain a general result.

Let the experimental spectrum μ be equal to the theoretical one scaled by $1-\epsilon$, with an additional noise peak in location $x \leq m_1$ with intensity ϵ (see Fig. 5). We will investigate which values of κ allow to correctly identify the additional peak as noise, and return the proportion of ν as $1-\epsilon$ instead of 1. To this end, we will compare the Wasserstein distance $W(\mu, \nu)$ with the cost of denoising κ. Note that, since the spectra are identical save for the noise peak, if this peak gets removed then κ is the full cost of regression (i.e. we get $W(\mu, p_0\omega + \nu_p) = \kappa$).

Let A_i be the cumulative intensity of the theoretical spectrum ν. The Wasserstein distance

$W(\mu,\nu)$ is equal to

$$
\begin{aligned}
W(\mu,\nu) &= \epsilon(m_1 - x) + \sum_{i=1}^{n-1} \frac{1}{q}\big|A_i - (1-\epsilon)A_i - \epsilon\big| \\
&= \epsilon(m_1 - x) + \sum_{i=1}^{n-1} \frac{1}{q}\big|\epsilon A_i - \epsilon\big| \\
&= \epsilon(m_1 - x) + \frac{\epsilon}{q}\sum_{i=1}^{n-1}(1 - A_i) \\
&= \epsilon(m_1 - x) + \frac{\epsilon}{q}(n - \sum_{i=1}^{n-1} A_i).
\end{aligned}
$$

Now, note that $\sum_{i=1}^{n-1} A_i = \sum_{i=1}^{n-1}(n-i)a_i = n(1-a_n) - \sum_{i=1}^{n-1} ia_i$, where the first equality comes from counting the number that each of the a_i variables occur in the sum, and the second one follows from the fact that the intensities sum to 1. This can be further simplified by noting that $n(1-a_n) - \sum_{i=1}^{n-1} ia_i = n - \sum_{i=1}^{n} ia_i$, where we included the term na_n in the sum.

Now, note that since the intensities are normalized,the average mass of the spectrum ν, denoted as $\bar{m}$, is equal to $\sum_{i=1}^{n}(m_1 + i/q)a_i$. This allows us to express $\sum_{i=1}^{n} ia_i$ simply as $q(\bar{m} - m_1)$, i.e. the distance between the average and the monoisotopic mass multiplied by the charge. Plugging this into the equation for $W(\mu, nu)$, we get a simple formula

$$W(\mu,\nu) = \epsilon(m_1 - x) + \epsilon(\bar{m} - m_1).$$

The cost of removing the noise peak will be lower than the cost of such transport whenever $\epsilon(m_1 - x) + \epsilon(\bar{m} - m_1) > \kappa$. It follows that the influence of the noise peak on the estimated proportions depends not only on it's proximity to the signal, $m_1 - x$, as is the case in methods based on linear regression, but also on the width of the isotopic envelope, $\bar{m} - m_1$. Note that the latter term is always positive for spectra with more than one peak. Therefore, the shape of the isotopic envelope facilitates the detection of noise peaks by the linear regression procedure based on the Wasserstein distance.

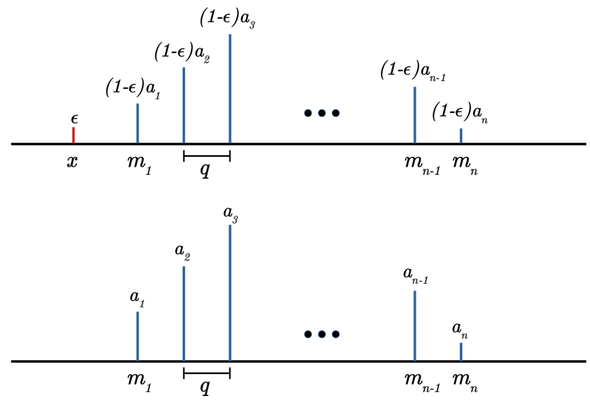

Figure 5: An example pair of experimental and theoretical spectrum, where the former (top) is equal to the latter (bottom) plus an additional noise peak.

The presence of the noise peak changes the proportion of ν from 1 to $1-\epsilon$, even though it is the only isotopic envelope in μ. This is because we estimate the amount of the total signal in μ that, under a given value of κ, can be feasibly explained by the model spectrum, and not the amount of ions. To obtain the latter, the proportions of the theoretical spectra returned by our method need to be normalized so that they sum up to 1.

Computation of the optimal proportions. The formulation of the regression problem discussed in the previous section is well suited for theoretical analysis. However, it does not show how to obtain the optimal proportions in practice. Below, we show an equivalent formulation that is better suited for implementation and practical applications. The proof that the two formulations are equivalent, based on Theorem 2, has been relegated to the Appendix.

Let $M(t)$, $N_j(t)$ be the cumulative distribution functions of μ and ν_j respectively. The cumulative distribution function of the model spectrum $\nu_p = \sum_{j=1}^{k} p_j\nu_j$ is equal to $N_p(t) = \sum_{j=1}^{k} p_jN_j(t)$.

Let $g(s_i)$ for $i = 1, 2, \ldots, n$ be the amount of μ's signal at the point s_i that is transported onto ω under a given transport plan. Note the distinction between g and ω: the latter is an auxilliary spectrum, therefore it's a concept analogous to μ and ν_p; the former denotes the amount of signal transported to ω, therefore being analogous to γ. Let $G(s_i) = g(s_1) + \cdots + g(s_i)$ be a cumulative distribution function of the unexplained signal. Note that $G(s_n) = g_{s_1} + \cdots + g_{s_n} = p_0$. Conceptually, G is a different construct than M or N, since it does not denote the amount of signal present in any given spectrum, but rather the amount of signal removed from μ by transporting it onto ω. In particular, we need to have $g(x) \leq \mu(x)$ for any point x.

For centroided spectra, the optimization problem (7) can be rephrased as a minimization over the variables p and g as follows:

$$p^*, g^* = \arg\min_{p,g} \left\{ \kappa p_0 + \sum_{i=1}^{n-1} (s_{i+1} - s_i) \Big| M(s_i) - G(s_i) - N_p(s_i) \Big| \right\}. \tag{8}$$

Note that the minimization above is performed over both p_j and $g(s_i)$ variables. Moreover, we require $p_0 + \sum_{j=1}^{k} p_j = \sum_{i=1}^{n} g(s_i) + \sum_{j=1}^{k} p_j = 1$, as all the signal in the observed spectrum needs to either be explained by the theoretical spectra or labeled as unexplained. The sum of p_j variables denotes the proportion of signal explained by the model spectrum.

Note that the optimization problem (8) is similar to formula (4) from Theorem 2, expressing the Wasserstein distance between a pair of spectra. It is equivalent to computing the Wasserstein distance between the model and the experimental spectrum after removing the unexplained signal from the latter, and additionally penalizing for the amount of signal removed. However, it does not give any clear physical interpretation of the denoising cost κ.

The minimization problem (8) is an example of the Least Absolute Deviation regression [19]. One of the common approaches to solving such problems is by using the technique of linear programming, which optimizes a linear function under a set of linear constraints [20]. However, the function to be minimized in problem (8) is not linear due to the absolute value, and the problem needs to be converted before using linear programming to solve it. There are several ways to convert a Least Absolute Deviation regression problem into a linear program, and choosing the proper one in each particular application is a crucial factor in obtaining a computationally efficient solution. We investigated several approaches and found that, in the case of linear regression of mass spectra, the approach based on the ideas described in [21] seems to be the most efficient.

The derivation of the linear program equivalent to the minimization problem (8) has been relegated to the Appendix. The main idea is to show that solving the regression problem is equivalent to solving the following linear program for the vector of variables z, and analyzing the differences between left- and right-hand sides of the inequalities:

$$\begin{array}{llll} \text{maximize} & V^T z & \text{over} & z \\ \text{subject to} & W^T z & \leq 0, & \\ & z_i - z_{i+1} & \leq & s_{i+1} - s_i, \quad i = 1, 2, \ldots, n-1, \\ & z_i - z_{i+1} & \geq & s_i - s_{i+1}, \quad i = 1, 2, \ldots, n-1, \\ & & & z \leq \kappa, \end{array} \tag{9}$$

Above, V is a vector of the observed spectrum intensities, such that $V_i = \mu(s_i)$, and W is a matrix of theoretical intensities, $W_{ij} = \nu_j(s_i)$. Note that V and the columns of W may represent spectra in either centroided or profile mode, because both types of spectra are represented as finite lists of m/z and intensity measurements when stored on computers. Moreover, since we only use the vectors of intensities and m/z values, we do not need to compute any additional values prior to solving the program (such as the cumulative sums used in the Wasserstein distance).

The minimization problem above can be solved using a number of algorithms for linear programming, such as the Simplex or the Interior Point methods. In our implementation, available in our Python 3 package `masserstein`, we have used the Simplex algorithm.

As in the case of computing the Wasserstein distance, the regression algorithm can be applied to profile experimental spectra provided that the sampling of intensity values is uniform over the m/z axis. This is how the results shown in Fig. 1 were obtained, with $\kappa = 0.02$. According to our knowledge, the Wasserstein distance is the first solution that allows for this kind of processing without the requirement for peak detection or centroiding. In the next section, we validate the performance of this approach on a set of experimentally obtained spectra as well as on simulated data.

Quality of experimental data. In real mass spectra, there are several different sources of unexpected signal, including chemical contaminants, random baseline electronic noise etc. In this work we do not distinguish between them, and broadly classify the experimental signal into two classes: the signal of the molecules of interest, referred to as the explained or expected signal, and all the other signal, referred to as the unexplained signal or noise.

Naturally, in noisy spectra, it is often far from clear whether a given signal is expected or not. Accordingly, any numerical method may mistake actual signal of interest for noise, and the other way around. Therefore, the signal transported from μ onto ω is the noise estimated by our method, which, as for any computational method, may differ from the actual noise signal.

One of the most difficult types of unexpected signal is the chemical contaminants with isotopic envelopes that highly overlap with the ones of the molecules of interest. In this case, the contaminating signal might be transported onto the theoretical envelope of the molecule of interest instead of ω. One of the possible ways of handling this problem is to use a database of common contaminants, such as the cRAP database[1], and include their theoretical isotopic envelopes in the model spectrum.

There are multiple factors other than the noise signal that influence the similarity of the observed spectra to their theoretical counterparts and the similarity between experimental spectra from replicate experiments. These include the variation of peak m/z position and the variation of peak intensities caused by random isotopologue sampling and measurement inaccuracies. Naturally, with any computational method, including ours, a change in the input will result in a change in the output. However, the variations in m/z and intensity values have different effects on the results obtained.

Due to the use of the optimal transport theory, our approach is inherently robust to the variation in the mass domain, as long as the parameter κ is properly adjusted and the variation is not excessive, as may be caused e.g. by an improper calibration of the instrument. Natural variations of m/z occurring in replicate experiments have close to no influence on the estimated proportions of molecules.

On the other hand, the variation in intensity has a pronounced effect on the obtained results, because it influences the amount of signal in the observed isotopic envelope of a given molecule. The variability of the signal intensity is therefore reflected in the variability of the estimated proportions. In particular, low ion counts lead to a highly unstable estimation. An example of this phenomenon is discussed in the Results section.

Although our method is fairly robust against measurement inaccuracies and sample impurities, any computational method will give improper results if the quality of the data is insufficient. Therefore, it is always the responsibility of the spectrometrist and the data analyst to first inspect the spectrum visually in order to determine its quality.

3 Results

3.1 Calibration mix study

In this section we verify whether the regression algorithm based on the Wasserstein metric can accurately filter out the background noise and estimate ion proportions in experimental spectra. We also compare the performance of the method on centroid and profile spectra.

As our test dataset, we take 200 repeated measurements of Pierce LTQ Velos ESI Positive Ion Calibration Solution (Thermo Scientific). This calibration mixture is composed of caffeine, a short

[1] https://www.thegpm.org/crap/

peptide with the sequence MRFA, and Ultramark 1621, a compound shown in Fig. 6. Note that due to varying side group lengths, Ultramark 1621 is in fact a mixture of 13 different compounds. In this dataset, all the isotopic envelopes of the compounds of interest are disjoint. We simulate the effect of overlapping envelopes by superposing shifted spectra.

Due to the different ionization rates of different compounds, it is difficult to predict the correct proportions of ion signals from their concentrations in samples. Therefore, we have assumed that our ground truth to which we compare our estimates are the true signal areas of compounds. To obtain the latter, we have manually selected the signal regions of all compounds, taking into account their monoisotopic peaks and peaks of isotopologues containing one additional neutron (i.e. first isotopic peaks). The region selection was performed on an average spectrum based on the 200 profile spectra. To ensure that the selected regions contain whole peaks in all the spectra, we have additionally taken into account the standard deviation of intensity at each point. The selected regions are shown in Supplementary Table S1, and a fragment of the average spectrum used to select them in Fig. 6. Next, for each spectrum we have integrated the signal within the selected regions using the trapezoidal rule. For each compound, the signals of its monoisotopic and first isotopic peaks were summed to yield the total signal of the compound.

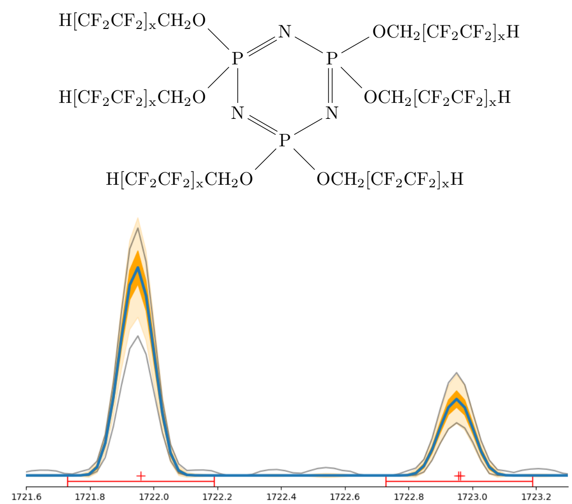

Figure 6: Top: Ultramark 1621, one of the compounds analyzed in this study; x=1,2,3. Bottom: A fragment of an average spectrum used to define m/z regions of ion signals, showing first two peak of an isotopic envelope of Ultramark 1621 with 14 CF_2CF_2 groups. Blue line shows the average signal. Dark orange and light orange ribbons show $\pm\sigma$ and $\pm 3\sigma$ regions, where σ is the standard deviation of signal intensity. Grey lines show the maximum and minimum signal over the 200 spectra. Red dots show the m/z values of theoretically predicted peaks. As the theoretically predicted masses agree well with the signal apexes, we infer that the spectrum is properly calibrated. As the maximum and minimum lines are in proximity of the $\pm 3\sigma$ ribbon, we infer that there are no outlying measurements of intensity. Random increases of maximum signal between the peaks indicate the presence of background noise.

We inspect the results of our Wasserstein regression method for several manually selected values of κ. The selection was based on the observed peak widths in profile spectra, which ranged from 0.013 at 195 Da to 0.38 at 1725 Da. In principle, setting κ to half base width of the broadest observed peak allows the method to use all the necessary experimental signal. However, we observe that the performance is best for $\kappa = 0.4$, i.e. the full base width of the broadest peak. This value allows for more flexible transport of signal between peaks of an isotopic envelope, effectively countering the effect of the variance of experimental peak intensity. The effect is mostly pronounced for low intensity peaks with small signal to noise ratio, in which case the particularly high variance of signal intensity has a detrimental effect on the estimation unless κ is sufficiently high. The reason behind this effect is explained in more detail in the following paragraphs.

Analysis of centroided spectra. The first goal of this study is to verify whether the Wasserstein regression algorithm implemented in the `masserstein` package returns accurate results when applied to a centroided experimental spectrum. In order to perform the centroiding in a controlled manner, we have implemented our own peak-picking procedure. Briefly, the spectra are centroided by integrating the signals within regions delimited by a fraction of 0.2 of the apex intensity. The

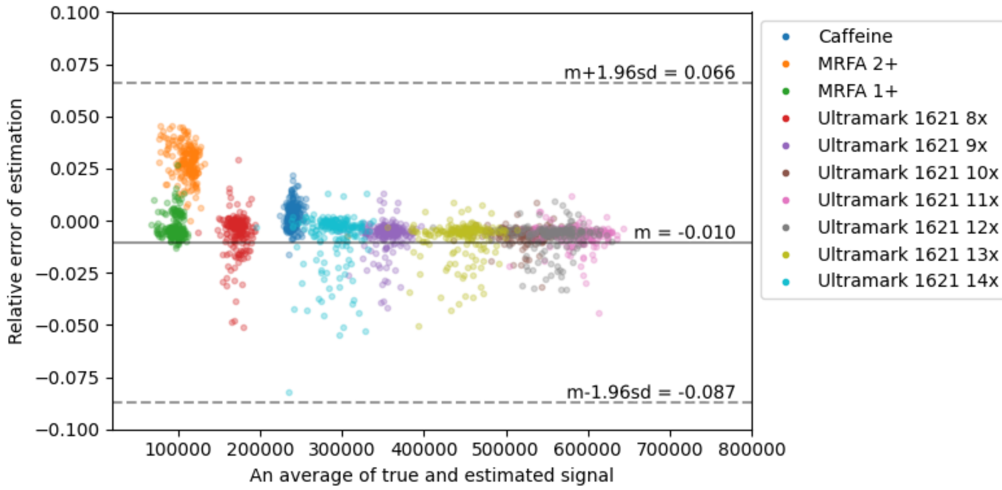

Figure 7: A Bland-Altman plot summarizing the Wasserstein regression results for the calibration mix on centroided spectra. Each point corresponds to an estimate of an ion intensity in one of the 200 spectra analyzed, with colors corresponding to different ions. The Y coordinate of each point corresponds to the estimated minus the true signal divided by the latter. Ultramark 1621 8x denotes Ultramark 1621 with 8 CF_2CF_2 groups, etc.

pseudo-code of the full procedure is shown in Algorithm 3 in section S4 of the Appendix. We have validated our implementation by comparing the centroided peak intensities with the manually integrated signal areas and found a good agreement.

We have generated the theoretical spectra of the compounds of interest using IsoSpecPy [22], assuming a proton adduct. In each spectrum we have retained the monoisotopic and the first isotopic peaks, so that the theoretical spectra correspond to the manually identified regions. After that, we have used `masserstein` to regress each of the 200 experimental spectra against our set of theoretical spectra. We have inspected several values of the denoising penalty κ ranging from 0.05 to 0.6. For $\kappa = 0.4$, the regression of all spectra took around 98 seconds on a single core of an Intel(R) Core(TM) i5-6300HQ CPU @ 2.30GHz processor

The `masserstein` package requires that all the spectra are normalized. Prior to normalization we have computed the total ion currents of the experimental spectra. After obtaining the estimated proportions of ions, they were multiplied by the total ion current of the experimental spectrum. This way we obtain the total signal (as opposed to the proportion of the total signal) explained by each ion, which we then compare to the manually calculated signal area.

The results for $\kappa = 0.4$ are shown in Fig. 7. The overall correlation between the estimated and the manually computed signal was equal to $\rho = 0.9998$. The mean difference between the estimated and manually integrated signal relative to the latter was equal to 1%, indicating a slight downward bias for this value of denoising penalty. The mean absolute relative difference was equal to 0.017, meaning that the average error of the estimation is equal to 1.7% of the true value. The detailed results for each ion are shown in Supplementary Fig. S1.

For $\kappa = 0.3$ the results were similar for most ions, and the overall correlation of the estimated and true signals was equal to $\rho = 0.9994$. However, we have found a strong down-estimation of the signal of Ultramark 1621 with 15 CF_2CF_2 side groups, which caused the drop in the correlation. We have found out that the bias is caused by a large variability of the relative peak intensities of this ion. In some spectra, the first isotopic peak was up to two times lower than on average. Such a large variability is most likely caused by a small number of ions of this compound.

Highly variable peak heights, combined with low denoising penalties, are detrimental to our current implementation of the Wasserstein regression. When the maximum feasible transport distance is low, the procedure necessarily fits to the smallest matching peak of the experimental spectrum. This is because, after all the signal of such peak is transported to a theoretical spectrum, there is no neighboring signal left that can be feasibly transported.

Increasing the denoising penalty allows to distribute the experimental signal more evenly over the theoretical spectrum, therefore increasing the accuracy of the estimation. However, when the penalty is too high, the background noise may also be transported to the theoretical spectrum, leading to an overestimation of an ion's signal.

The results demonstrate that the optimal transport theory applied to the problem of linear regression of mass spectra is capable of giving very accurate estimates of ion signals, provided that all the considered signals are over the limit of quantification to ensure small variability of peak areas. As we show below, it also opens the possibility of directly analyzing the profile spectra without the peak-picking step.

Analysis of profile spectra. In the current implementation of the Wasserstein regression algorithm, we treat profile spectra in the same way as the centroided ones. As discussed in the previous section, this approach gives a good approximation when the signal sampling is uniform over the m/z axis. However, this is often not the case for spectrometers which have a non-constant resolving power. As peaks get broader with the increasing m/z value, less data points are needed to reflect the signal shape. This phenomenon is often exploited in order to decrease the data size. One of the way to circumvent this problem in order to use the current implementation of our method is to resample the signal intensities.

We have resampled our spectra using a piecewise linear interpolation, in which the signal intensity in each point is approximated by a weighted average of the neighbouring intensities. The full procedure is shown in pseudo-code in Algorithm 2 in section S4 of the Appendix. For each spectrum, we have interpolated its signal such that the spacing between neighboring m/z values was 0.001.

The downside of the resampling strategy is a large increase of the data complexity and, consequently, the computational time. For the regression of the 200 profile spectra with $\kappa = 0.4$, the computations took 35 minutes, compared to 98 seconds for centroided spectra.

The penalty $\kappa = 0.4$ yielded an overall correlation of 0.9998. The results for all compounds are shown jointly in Fig. 7, and for each ion in detail in Supplementary Fig. S2. We have noticed a systematic slight overestimation of the caffeine signal, most likely due to incorporation of a small unidentified peak at 195.72 Da which was present in most spectra. For $\kappa = 0.3$ we have observed a bias in the estimation of Ultramark 1621 with 15 CF_2CF_2 groups, similar as for the centroided spectra.

Further information about the results of the regression can be obtained by inspecting the spectrum of the remaining signal (i.e. the signal not explained by the theoretical spectra). An example of such spectrum, obtained for one of the 200 mass spectra, is shown in Fig. 8. The remaining signal corresponds to approximately 45% of the total ion current. By inspecting the spectrum of the remaining signal we conclude that all the molecules of interest were properly detected. However, not all of the signal of interest was used for regression, most likely due to the peak height variability discussed in the previous paragraph. We also detect contaminating ions, one of which is visible at 1395 Da in the right panel of Fig. 8, and numerous apparently random peaks. On the other hand, inspecting the fragment with caffeine (not shown) confirmed our assumption that a contaminant with a highly overlapping isotopic envelope caused an overestimation of the signal. Further research into the application of optimal transport to the processing of mass spectra should allow for a better separation of the signal of interest from the contaminants.

In general, the results were similar to the ones obtained on centroided spectra. This shows that `masserstein` allows for the processing of profile spectra without the need of peak peaking. Further research into applications of the optimal transport theory to the processing of mass spectra has the potential to increase both the computational efficiency and the accuracy of the results.

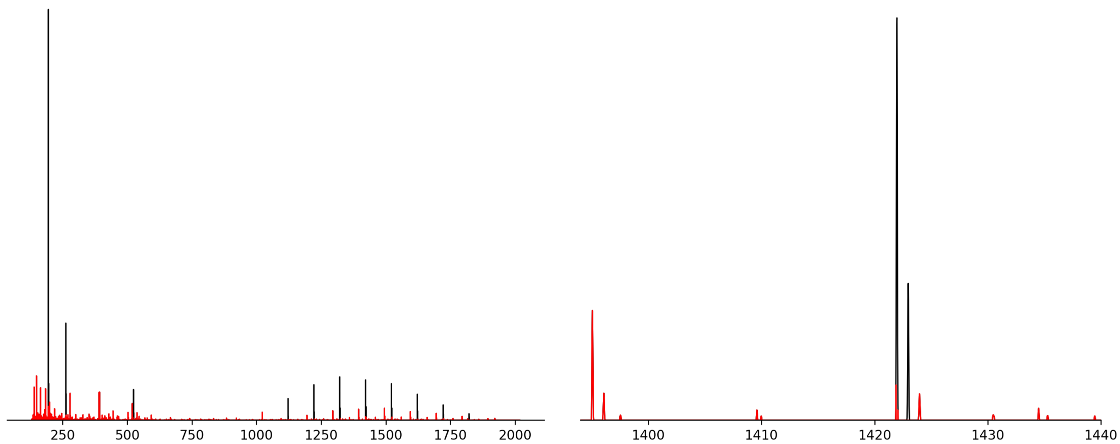

Figure 8: One of the 200 mass spectra used in this study with the signal remaining after regression with $\kappa = 0.4$ highlighted in red. Left: Full spectrum. Black peaks, corresponding to no remaining signal, indicate that the molecules of interest were properly detected. Right: A zoomed in fragment containing the isotopic envelope of Ultramark with 11 CF_2CF_2 groups at 1421.77 Da. The signal from the first isotopic peak was fully used for regression, while there is still some remaining signal of the monoisotopic peak. As expected, the signal from the second isotopic peak was not used, because this peak was discarded from the theoretical spectrum.

Overlapping isotopic envelopes. In our final experiment on the calibration mix data, we investigate the influence of overlapping envelopes on the accuracy of the estimation. In the case of disjoint isotopic envelopes, like in the previous experiments, it is easy to obtain the ground truth by manually selecting peak regions for integration. However, the task is complicated when the peaks of the envelopes overlap, because manual integration does not allow to separate them and compute their individual signals. Therefore, we have decided to simulate this effect using the calibration mix data.

For each spectrum in profile mode, we have created its copy shifted by one hydrogen mass. Each spectrum was mixed with its shifted copy in proportion 0.7 of the original and 0.3 of the copy. The spectra were subsequently centroided as in the previous examples.

To generate a model spectrum, we have taken the formulas from the previous experiments, and the same formulas with one additional hydrogen. When adding a hydrogen atom to the sum formula, we only modified the ion's mass, while keeping its charge unchanged. Note that, from a computational perspective, it does not matter whether the formulas obtained this way correspond to any actual chemical compound.

From the computational point of view, this dataset is much more difficult than the previous ones, and we expect a decrease in estimation accuracy. One of the reasons is that when overlapping signals merge, the apex positions shift relative to the original ones. Therefore, peak picking a profile spectrum with overlapping signals returns distorted peak positions, and for a sufficiently large overlap one gets a single peak instead of two. This poses a major difficulty for approaches based on pointwise comparison of spectra, while the Wasserstein distance is more robust to such changes.

Using `masserstein`, we have regressed the mixed spectra with denoising penalty $\kappa = 0.4$. This time, the regression of all the spectra took approximately 3 minutes (not including the time needed for centroiding). The results were compared with the signal areas integrated in the previous experiment, rescaled either by 0.7 or 0.3 to accommodate for the mixing proportions. The correlation between the estimates of `masserstein` and the true signals was equal to $\rho = 0.9985$, only slightly smaller than for the previous datasets. The detailed results are shown in Supplementary Fig. S3.

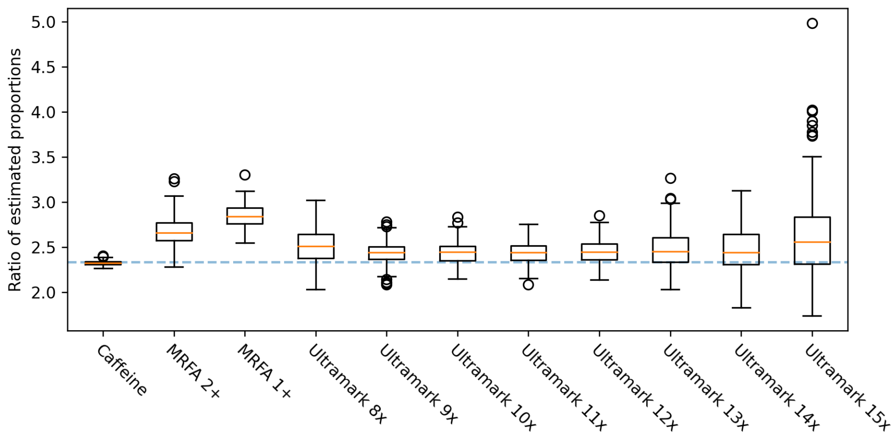

Figure 9: The ratio of estimated signals of ions with overlapping isotopic envelopes. Each boxplot corresponds to a pair of ions with sum formulas differing by one hydrogen atom. The dashed line represents the true ratio.

Additionally, we have calculated the ratios of estimated proportions of corresponding ions. In each spectrum we have compared the estimated signal of an original ion to the estimated signal of it's counterpart with one additional hydrogen atom. We have compared the ratio obtained this way to the reference value of $0.7/0.3 \approx 2.33$. The result is shown in Fig. 9.

We have observed that the ratio is overestimated for MRFA 1+. Comparing the ratio with the results shown in Supplementary Fig. S3, we conclude that this is caused by an overestimation of the signal of the original MRFA 1+ ion (i.e. without the added hydrogen atom). For all the other ions, the true ratio was within the 95% confidence interval of the estimation. The estimated ratio showed a high variance for Ultramark 15x, which is likely caused by the lower signal to noise ratio of this ion compared to the other ions. On the other hand, the estimated signal ratio was the most accurate for caffeine, likely due to low amount of background noise in the neighbourhood of its isotopic envelope and a sufficiently high signal intensity.

The results of the three experiments presented in this subsection show that our Wasserstein regression algorithm implemented in `masserstein` is capable of accurate estimation of signal intensity based on experimental spectra in both centroided and profile mode, and also in the presence of overlapping isotopic envelopes. However, the dataset presented in this section contained only a handful of molecular formulas. In order to verify the results for a broader range of molecules, and to demonstrate that `masserstein` is not limited to any particular type of ions (be it lipids, peptides or metabolites), in the next subsection we perform an extensive analysis on simulated datasets consisting of purely random molecular formulas.

3.2 Computational experiments

In this section, we evaluate the accuracy and bias of the estimation performed by `masserstein` using simulated mass spectra. We introduce a series of measurement distortions into the observed spectrum in order to reflect the variability of peak heights and limited resolving power and accuracy observed in real spectra.

We have created several simulated datasets by computing spectra of mixtures of randomly generated molecules. In each dataset, all the molecules had the same nominal mass in order to ensure high overlap of their isotopic envelopes. The molecules were simulated by subsequently sampling elements in the order of C, O, N, S, and P. Note that any user-supplied formula can be used in `masserstein`, as no mathematical procedure used in this work depends on the chemical properties of molecules. Therefore, in order to simplify the simulation procedure, we did not restrict the sampled molecules to ones which are chemically possible.

We have simulated datasets consisting of isobaric molecular formulas for a range of nominal masses from 60 to 12 000 Da and from 1 to 8 isotopic envelopes. Based on the formulas, we have generated theoretical spectra using IsoSpecPy [22]. To obtain simulated experimental spectra, for each dataset we have mixed the theoretical ones in random proportions. We have generated both centroid and profile experimental spectra. In the latter case, we have assumed a Gaussian shape of peaks.

To each experimental spectrum, in both profile and centroid mode, we have introduced extensive distortions to simulate the effects of a finite number of molecules, electronic noise, measurement inaccuracy in mass and intensity domain and limited resolving power (100 000 at 600 Da in the case of profile spectra). The procedure is described in detail in section S3 of the Appendix.

To assess the quality of regression results, we calculate the mean absolute deviation (MAD) between true and estimated signal contributions of the theoretical spectra:

$$\mathrm{MAD} = \frac{1}{k}\sum_{i=1}^{k} |\hat{p}_{-} p_i|$$

In the above formula, k is the number of theoretical isotopic envelopes, p_i is the true proportion of the i-th envelope, and $\hat{p}_i$ is its estimated proportion. We do not directly compare the amount of unexplained signal, p_0, as this information is implicitly included in the sum of the estimated proportions.

In the case of profile observed spectra, we have inspected two denoising penalties, equal 0.0075 (half peak base width) and 0.08. The results are shown in Supplementary Fig. S5. For the lower penalty, the MAD was mostly between 10^{-2} and 10^{-1}, indicating at least one accurate decimal digit in an average estimate. We have observed some bias in the estimation, as the estimated proportions were usually lower than the true ones. The mean of the residues was equal to -0.0197, while their standard deviation was 0.025. The bias increased with increasing true proportion. This can be explained by the fact that the isotopic envelopes of abundant ions have a larger variance of their peak heights due to isotopologue sampling. For the 0.08 penalty, the MAD was smaller and usually around 10^{-2}. The decrease in MAD was especially pronounced for low numbers of isotopic envelopes. One of the reasons for better performance in this case is a lower bias, as the mean of the residues was -0.008. However, at the same time, their variance increased to 0.028. In all cases, the processing of a single spectrum took less than one second on a laptop computer with Intel(R) Core(TM) i5-6300HQ CPU @ 2.30GHz processor.

In the case of centroided spectra, the denoising penalty was set to 0.02 (ten times the standard deviation of the simulated m/z measurement error). The results are shown in Supplementary Fig. S4. The MAD was mostly between 10^{-3} to 10^{-2}, indicating at least two accurate decimal digits an average estimation. There were no clear differences between low and large masses, indicating that isotopic envelopes of 200 Da ions have enough information for accurate regression using our method. In most cases, the best estimates per observed spectrum had at least three accurate decimal digits. We did not observe any bias of the estimation. The mean of the residues was one order of magnitude lower than in the case of profile spectra.

The results show that our method is able to perform accurate estimation of molecule proportions even in the case of several isobaric interferences and additional chemical noise. Moreover, it is applicable to both profile and centroided spectra. In the case of profile spectra, we have observed a tradeoff between the estimation bias and variance for different values of the denoising penalty κ. The results presented above were better for centroided spectra. However, the two different types of distortions used to simulate both types of observed spectra are not directly comparable, and their magnitudes differ.

Notably, we have obtained accurate results for up to 6 isobaric molecules of 200 Da in the presence of 50 additional interfering peaks, even though the molecules themselves have only a few peaks in their isotopic envelopes. In this case, the largest absolute error of estimation per centroided spectrum, averaged over 100 replicates, was 0.026 (see Supplementary Fig. S4). This indicates that, on average, all the estimates had at least one accurate decimal digit.

4 Conclusions and discussion

In this work, we present advances in theoretical studies on the problem of linear regression of mass spectra, defined as fitting a linear combination of theoretical spectra to an experimental one. The problem is ubiquitous in mass spectrometry, appearing either implicitly or explicitly in areas as diverse as metabolomics, proteomics and polymer science. The theoretical foundations of the methods studied in this work are not limited to any particular type of experiment. Furthermore, the method can be readily applied to other types of spectra, such as the nuclear magnetic resonance ones, as long as reliable reference spectra are available. The broad range of applications is achieved by focusing on an abstract problem of fitting a linear combination of reference signals to an experimentally measured one, as opposed to developing a method restricted to a particular application.

One of the major factors that hinder the performance of currently available linear regression methods is the fact that the locations of experimentally measured peaks never match the theoretically predicted ones perfectly. This is caused, among others, by measurement inaccuracies, which, even if seemingly small, are unavoidable. Although they may be negligible when the spectra are analyzed manually, an m/z difference as small as a millionth of a Dalton means that a computer program sees the peak locations as different.

In order to circumvent this problem, the currently available software matches peaks within so called mass windows of predefined width [1, 4]. The width needs to be specified by the user, which makes it more difficult to apply this kind of approach in practice. The choice of a mass window width is further complicated by the occurrence of highly overlapping peaks in profile spectra. In some cases, the individual apexes of such peaks are no longer visible. Instead, we obtain a single apex located between the "true" apexes of the component peaks. This kind of merging of peaks, occurring especially in complex profile spectra, leads to an increased difference between theoretical and observed peak locations in centroided spectra. Therefore, especially in complex or low-resolution spectra, the optimal window width may be considerably greater than the nominal accuracy of the instrument.

Even if the user knows the optimal width of a mass window, this kind of approach has several intrinsic drawbacks. Due to the infinite resolution of a theoretically predicted spectrum, a mass window usually contains several theoretical peaks. Within the window, those peaks are effectively treated as one. Therefore, this procedure effectively limits the resolution of the *in silico* predicted spectrum, leading to an unnecessary loss of information. On the other hand, it tends to merge closely positioned signal and noise peaks in the experimental spectrum, which influences the estimated proportions.

In order to alleviate those difficulties, we have investigated the application of the Wasserstein distance to the problem of linear regression of mass spectra. Methods based on quantifying the distance in the m/z domain needed to transform one spectrum into the other are naturally robust to limited resolving power and accuracy of instruments. This robustness makes them a promising tool for methods based on comparing experimentally acquired spectra.

The pre-existing, extensive mathematical research on the topic of optimal transport has resulted in powerful theorems that allow to express the Wasserstein distance as a computationally feasible integral. Those theorems have allowed us, in turn, to develop an effective algorithm for Wasserstein regression that does not require any optimization of the transport plan. Instead, the optimal cost of transporting the signal between spectra is computed in a straightforward manner using the aforementioned theorems, and a technique of linear programming is used to directly optimize the proportions of theoretical spectra.

The choice of the denoising penalty. Our method requires the user to specify a single parameter: the cost of denoising κ, interpreted as the maximum feasible transport distance in the m/z domain. This interpretation of κ allows to treat it analogously to the radius of a mass window. However, a major difference between the two approaches is that κ acts as a *soft threshold*: during regression, whether some signal is transported between two peaks depends not only on their distance relative to κ, but also on the shape of the theoretical isotopic envelopes. This allows for some flexibility in setting the value of κ and makes the results more stable when the value is sub-optimal.

Similarly to the width of a mass window, in practical applications the choice of κ is not straightforward. The performance of the regression methods for a given parameter value is influenced by several factors, including the instrument accuracy and resolving power. Usually, the results for several different values need to be inspected manually in order to make a final decision. The half-base widths of peaks in profile spectra and the instrument accuracy serve as a convenient reference for the reasonable range of values of this parameters. Setting κ higher than 1 is usually inadvisable, as it allows to transport the signal between peaks of ions with different atomic compositions.

The choice of κ resembles the classical variance-bias trade-off known from the field of machine learning. Small values of κ lead to down-estimation of ion proportions, as insufficient signal is available to be transported onto their theoretical isotopic envelopes. On the other hand, while high values of κ allow to transport all the required signal, they lead to incorporation of noise in the estimated proportions, therefore increasing the variance of the estimation. Additionally, if systematic sample impurities are present, high values of κ may lead to an over-estimation of the amounts of the molecules of interest. Further studies are needed in order to give precise guidance as to the optimal value of this parameter.

Further research. The Wasserstein distance quantifies the similarity of spectra only in terms of the difference in m/z locations of signals. A drawback of this approach is that it is relatively sensitive to differences in peak intensities. This is further emphasized by the different nature of measurement uncertainties in the mass and the intensity domain, caused by the fact that mass spectrometers measure the m/z and the number of ions differently. Although accounting for this phenomenon would be desirable, it is highly non-trivial to formalize it mathematically in a way that would be suitable for applications in mass spectrometry and linear regression of spectra in particular. Especially in the latter, allowing peak intensities to vary may lead to instability of the results—when everything is variable, we can fit anything to anything. For this reason, in this work we only allow the signal to be removed from the experimental spectra, and not from the theoretical ones.

A practical consequence of allowing the signal to only be removed from the experimental spectrum is that, when simulating the theoretical spectra, we need to discard peaks that are under the level of quantification. The measured intensities of such peaks are unreliable, leading to erroneous results for the whole ion. In an extreme case, when a peak is missing in the experimental spectrum but is present in the theoretical one, it forces the estimated proportion to be zero. The lack of experimental intensity that can be transported onto this theoretical peak means that the whole isotopic envelope needs to be discarded. Allowing for some flexibility in the intensities of theoretical peaks would allow to alleviate this difficulty. However, in order to accurately reflect the observed variability of peak intensities, the regression procedure needs to be coupled with a mathematical model of the shape of experimentally measured isotopic envelopes. Whether such coupling is mathematically and computationally feasible remains an open question.

Implementation. We have implemented the discussed algorithms in a Python 3 package called `masserstein`. Our implementation is designed to be applicable in larger data processing pipelines. Efficient development of pipelines requires freely available modular tools, which perform specific tasks and can be easily combined. Therefore, our implementation does not perform any additional pre- or postprocessing of the results, such as peak-picking or correcting for proton affinity of molecules. Such procedures can be performed separately using designated tools, available e.g. in the OpenMS package [23].

On a final note, we reiterate that, from a methodological point of view, molecule identification and quantification are two separate tasks. Accordingly, in this work, we assume that the chemical formulas of the molecules to be quantified are known a priori. These may come either from the scientific question at hand (such as the frequency of a given posttranslational protein modification), from the knowledge about the experimental setup (such as whether lipid extraction was performed), or from an identification study performed prior to quantitative analysis (e.g. using the SIRIUS program [24]). Another approach, applicable to proteomics, is to use the averagine model of amino acid, as exemplified by the MasSPIKE program [25].

Acknowledgements. This work has been supported by the Special Research Fund (BOF) of the Hasselt University BOF18BL09, the FWO-PAS grant VS.028.19N, and the Polish National Science Centre grants #2018/29/B/ST6/00681 and #2017/26/D/ST6/00304.

References

[1] Lermyte, F. et al. "Understanding reaction pathways in top-down ETD by dissecting isotope distributions: A mammoth task". In: *International Journal of Mass Spectrometry* 390 (2015), pp. 146–154.

[2] De Bruycker, K., Krappitz, T., and Barner-Kowollik, C. "High Performance Quantification of Complex High Resolution Polymer Mass Spectra". In: *ACS Macro Letters* 7.12 (2018), pp. 1443–1447.

[3] Engler, M. S. et al. "COCONUT - An Efficient Tool for Estimating Copolymer Compositions from Mass Spectra". In: *Analytical chemistry* 87.10 (2015), pp. 5223–5231.

[4] Peckner, R. et al. "Specter: linear deconvolution for targeted analysis of data-independent acquisition mass spectrometry proteomics". In: *Nature methods* 15.5 (2018), p. 371.

[5] Meija, J. and Caruso, J. A. "Deconvolution of isobaric interferences in mass spectra". In: *Journal of the American Society for Mass Spectrometry* 15.5 (2004), pp. 654–658.

[6] Meija, J., Beck, T. L., and Caruso, J. A. "Interpretation of alkyl diselenide and selenosulfenate mass spectra". In: *Journal of the American Society for Mass Spectrometry* 15.9 (2004), pp. 1325–1332.

[7] Dasari, S. et al. "Quantification of isotopically overlapping deamidated and 18o-labeled peptides using isotopic envelope mixture modeling". In: *Journal of proteome research* 8.3 (2009), pp. 1263–1270.

[8] Ferrige, A. et al. "Maximum entropy deconvolution in electrospray mass spectrometry". In: *Rapid communications in mass spectrometry* 5.8 (1991), pp. 374–377.

[9] Lacki, M. K. et al. "masstodon: A Tool for Assigning Peaks and Modeling Electron Transfer Reactions in Top-Down Mass Spectrometry". In: *Analytical chemistry* 91.3 (2019), pp. 1801–1807.

[10] Horn, D. M., Zubarev, R. A., and McLafferty, F. W. "Automated reduction and interpretation of". In: *Journal of the American Society for Mass Spectrometry* 11.4 (2000), pp. 320–332.

[11] Du, P., Kibbe, W. A., and Lin, S. M. "Improved peak detection in mass spectrum by incorporating continuous wavelet transform-based pattern matching". In: *Bioinformatics* 22.17 (July 2006), pp. 2059–2065. ISSN: 1367-4803. eprint: `http://oup.prod.sis.lan/bioinformatics/article-pdf/22/17/2059/767773/btl355.pdf`.

[12] Wan, K. X., Vidavsky, I., and Gross, M. L. "Comparing similar spectra: from similarity index to spectral contrast angle". In: *Journal of the American Society for Mass Spectrometry* 13.1 (2002), pp. 85–88.

[13] Reinhold, B. B. and Reinhold, V. N. "Electrospray ionization mass spectrometry: Deconvolution by an entropy-based algorithm". In: *Journal of the American Society for Mass Spectrometry* 3.3 (1992), pp. 207–215.

[14] Majewski, S. et al. "The Wasserstein Distance as a Dissimilarity Measure for Mass Spectra with Application to Spectral Deconvolution". In: *LIPIcs-Leibniz International Proceedings in Informatics*. Vol. 113. Schloss Dagstuhl-Leibniz-Zentrum fuer Informatik. 2018.

[15] Villani, C. *Optimal transport: old and new*. Vol. 338. Springer Science & Business Media, 2008.

[16] Rubner, Y., Tomasi, C., and Guibas, L. J. "The earth mover's distance as a metric for image retrieval". In: *International journal of computer vision* 40.2 (2000), pp. 99–121.

[17] Jablonski, J. and Marciniak-Czochra, A. "Efficient algorithms computing distances between Radon measures on R". In: *arXiv preprint arXiv:1304.3501* (2013).

[18] Santambrogio, F. *Optimal transport for applied mathematicians*. Springer, 2015, pp. 99–102.

[19] "Least Absolute Deviation Regression". In: *The Concise Encyclopedia of Statistics*. New York, NY: Springer New York, 2008, pp. 299–302. ISBN: 978-0-387-32833-1.

[20] Vanderbei, R. J. et al. *Linear programming*. Springer, 2015.

[21] Wagner, H. M. "Linear programming techniques for regression analysis". In: *Journal of the American Statistical Association* 54.285 (1959), pp. 206–212.

[22] Łacki, M. K. et al. "IsoSpec: Hyperfast Fine Structure Calculator". In: *Analytical chemistry* 89.6 (2017), pp. 3272–3277.

[23] Röst, H. L. et al. "OpenMS: a flexible open-source software platform for mass spectrometry data analysis". In: *Nature methods* 13.9 (2016), p. 741.

[24] Dührkop, K. et al. "SIRIUS 4: a rapid tool for turning tandem mass spectra into metabolite structure information". In: *Nature methods* 16.4 (2019), p. 299.

[25] Kaur, P. and OConnor, P. B. "Algorithms for automatic interpretation of high resolution mass spectra". In: *Journal of the American Society for Mass Spectrometry* 17.3 (2006), pp. 459–468.